

C++23, program analysis, and low-level engineering: the PS-form Analyzer, AdvancedAlgorithms, an LLVM-track internship, NASM IA-32, and x86-64 codegen.

1st · 104/104

PS-form; only maximum result among 49

12 C++23 modules

contracts, differential tests, sanitizers, stress

500k vertices

stress gates catch recursion and accidental $O(n^2)$

EXPERIENCE

- JULY 1 —
AUGUST 31, 2026

MCST · LLVM · 0.25 FTE

 - 0.25 FTE: graph algorithms, compiler construction, and LLVM.
 - Developing an LLVM optimization pass.
- 2026 — PRESENT

AdvancedAlgorithms

 - 12 reusable C++23 modules with explicit complexity and validation.
 - Large tests up to 500k vertices catch recursion and accidental $O(n^2)$.

SELECTED PROJECTS

- PS-form Analyzer

Conservative memory-dependence analysis; ranked 1st of 49 with 104/104.
- x86-64 Codegen Lab

SSA-like IR, liveness, linear scan, iced-x86, and isolated execution.
- UniversalToolchain/Wist2

Typed routes, callable-first SSA, and experimental PlanFuzz; 1,465/1,465 canonical gate.

SKILLS

- C++ systems

C++23, CMake, contracts, sanitizers, static analysis
- Program analysis

graphs, data flow, memory dependence, liveness
- Low-level

NASM IA-32, CDECL, x87, x86-64 codegen
- Languages

C++23, C17, C#/.NET, Python, Rust

EDUCATION

Software Engineering student at HSE University

RECOGNITION

HSE Vysshaya Proba prize-winner; top MEPhI Junior results in 2025/2026; Baltic Grand Prize.

TECHNICAL COMMUNICATION

Code review, documentation, and a practical NASM IA-32 course.

AVAILABILITY

Moscow / remote · up to 20 hours/week from September 2026